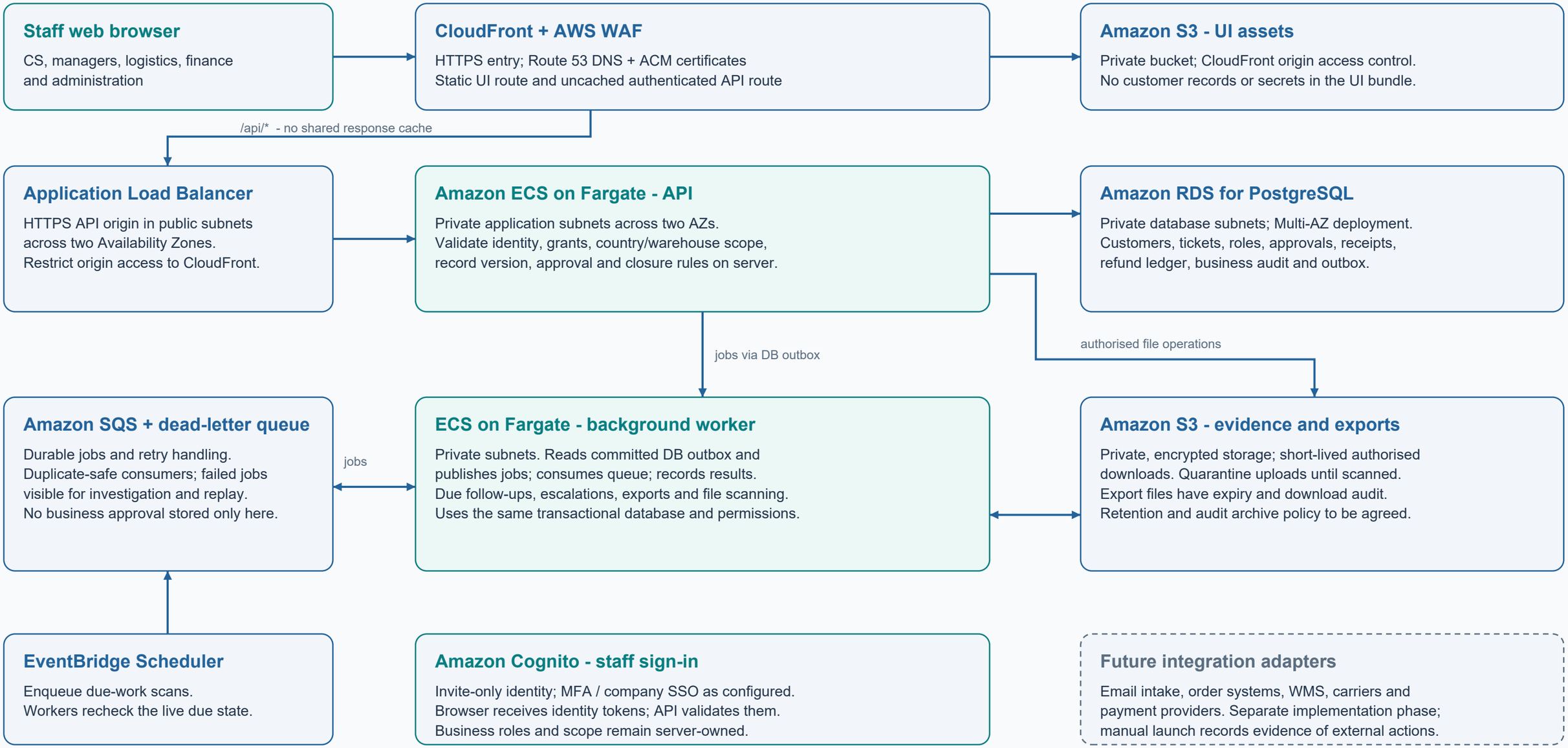
SUPPORT DESK / VERSION 6

Proposed AWS deployment architecture

Logical service view. V6 remains a review prototype; production infrastructure and integrations have not been deployed.



CONTROLS ACROSS ALL COMPONENTS

IAM task roles and least privilege; KMS encryption; Secrets Manager; CloudWatch logs, metrics and alarms; CloudTrail infrastructure audit; tested database backups and restore; separate dev, staging and production; infrastructure as code and gated deployment. Business audit is application-owned and is not replaced by CloudTrail.

Proposed AWS deployment architecture

Implementation notes and decisions. This maps the V6 specification to AWS services; final sizing, region and budget remain open.

1. Recommended starting shape

Use one modular application API and a separately scalable worker, with a relational database for transactional records. This keeps cross-module approvals, receipts, money and closure consistent without requiring a microservice estate at launch. Run production API capacity across two AZs; agree worker redundancy and scaling against the availability target. This is a recommendation, not a measured capacity plan.

2. Identity, permissions and access

Use Cognito or federated company SSO for authenticated staff identities. Disable public self-registration. The API must verify tokens and recheck active account state, grants, explicit approval authority and country/warehouse scope. The prototype role selector must not ship to production. Staff have no direct database access. The diagram omits token exchange arrows for readability; browser sign-in and API token validation are required.

3. Network and file boundaries

Only the HTTPS entry and load balancer are public. API, worker and database use private subnets. Restrict the load balancer origin to CloudFront traffic and validate an origin secret/header; configure security groups and TLS between tiers (HTTPS for HTTP traffic). Use VPC endpoints where applicable and controlled egress/NAT for other dependencies. S3 evidence stays separate from public UI assets; allow access only after authorisation and malware checks.

4. Transaction, follow-up and closure safety

Commit each business change, immutable audit event and outbox record in one database transaction. The worker publishes committed outbox events and retries safely; SQS delivery is not proof of completion. Apply idempotency keys, record versions and per-request concurrency checks. Closure must re-evaluate current approvals, outstanding return/waiver quantities, inspections, shipment delivery, finance and customer follow-up obligations. A return waiver must never overwrite a physical receipt.

5. Analytics, exports and manual launch

Start with indexed SQL and controlled reporting queries over the approved metric definitions. Add reporting projections or a read replica only when workload testing justifies them; never use stale analytics as the authority for financial or closure decisions. Queue larger CSV exports and store them privately with expiry, download audit and current access checks. At launch, staff record external communications, shipments and payments manually with evidence; no connector is implied to be active.

6. Delivery and operational acceptance

Use infrastructure as code and separate environments; build immutable images in CI and store them in Amazon ECR. Gate releases on role and scope tests, concurrent approval/receipt/refund tests, retry tests, upload security, export access, backup restore and end-to-end business acceptance. Observe overdue jobs, failed queues, API errors and business exceptions. CloudTrail covers AWS activity; application audit must retain actor, exact revision, changes and reason with tamper-resistant retention controls.

7. Decisions before provisioning

Confirm AWS account ownership, region/data residency, SSO/MFA policy, business SLA and technical availability target, recovery objectives, retention periods, expected staff/records/files, integration scope and monthly budget. Choose instance/task sizes, backup retention, network egress and scan tooling from those decisions. Multi-AZ is high availability, not a substitute for backup or a tested disaster-recovery plan. No cost estimate or regulatory compliance claim is made here.

AWS references consulted

- ECS load balancing
- Cognito user pools
- RDS Multi-AZ instances
- CloudFront origin controls
- Transactional outbox
- EventBridge Scheduler

Functional authority: Support Desk V6 Development Specification, 30 October 2026. AWS services are proposed implementation choices.